

Computational Microelectronics

Lecture 3 (Pre-recorded)

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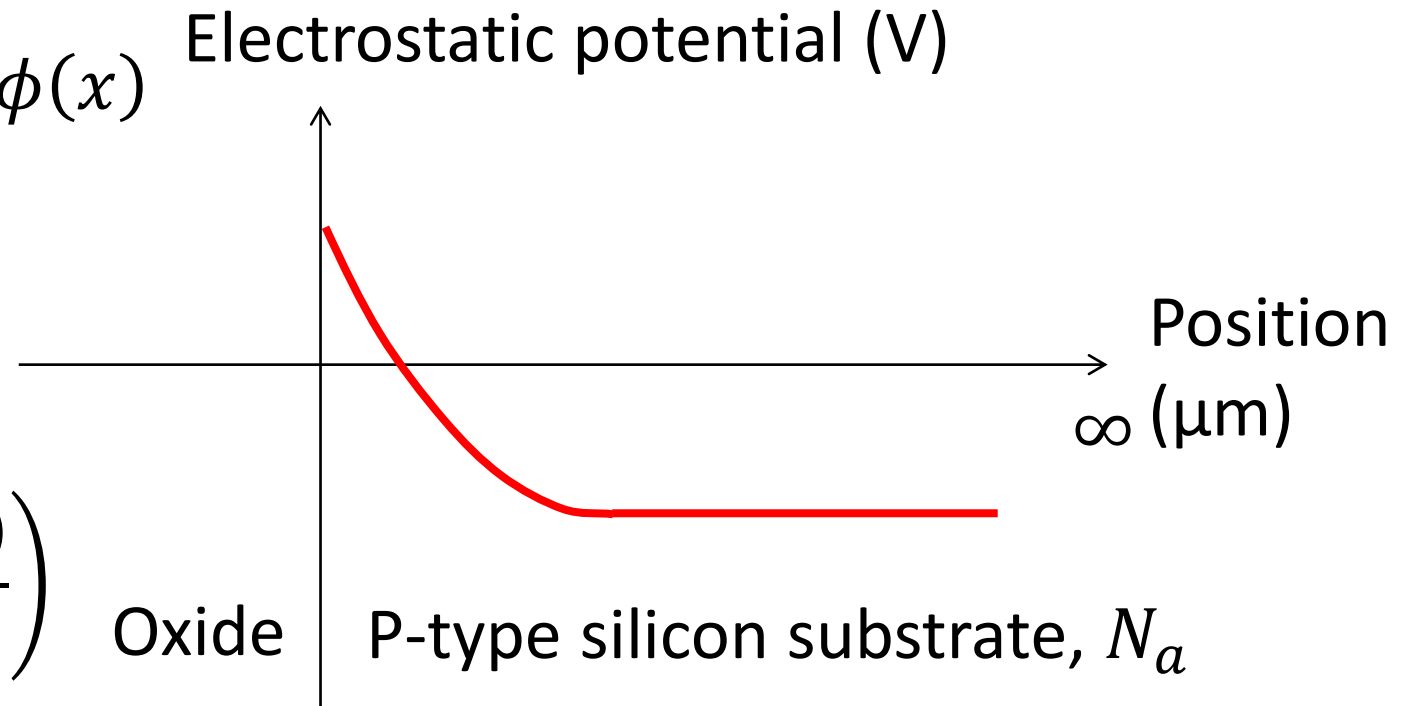
L3

MOS capacitor

- Problem specification
 - P-type silicon substrate, N_a
 - Interface at $x = 0$
 - Semi-infinite substrate
 - Electrostatic potential, $\phi(x)$

$$n(x) = n_{int} \exp\left(\frac{q\phi(x)}{k_B T}\right)$$

$$p(x) = n_{int} \exp\left(-\frac{q\phi(x)}{k_B T}\right)$$



Quantities of interest

- We want to know

- The electrostatic potential at the interface, $\phi(x = 0)$
- More precisely,

$$\phi_s = \phi(x = 0) - \phi(x = \infty)$$

- At $x = \infty$,

- The electrostatic potential at the substrate, $\phi(x = \infty) = \phi_\infty$, satisfies

$$N_a = n_{int} \exp\left(-\frac{q}{k_B T} \phi_\infty\right) - n_{int} \exp\left(\frac{q}{k_B T} \phi_\infty\right)$$

- The electric field vanishes at the substrate, $\left.\frac{d\phi}{dx}\right|_\infty = 0$

(Modified) Poisson equation

- Poisson equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}} [p(x) - n(x) - N_a]$$

– Multiplying the both sides by $\frac{d\phi}{dx} dx$, we have

$$\frac{d\phi}{dx} d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}} \left[n_{int} \exp\left(-\frac{q\phi}{k_B T}\right) - n_{int} \exp\left(\frac{q\phi}{k_B T}\right) - N_a \right] d\phi$$

– By using an expression for N_a ,

$$\begin{aligned} \frac{d\phi}{dx} d\left(\frac{d\phi}{dx}\right) &= -\frac{q}{\epsilon_{si}} n_{int} \\ &\times \left[\exp\left(-\frac{q\phi}{k_B T}\right) - \exp\left(\frac{q\phi}{k_B T}\right) - \exp\left(-\frac{q\phi_{\infty}}{k_B T}\right) + \exp\left(\frac{q\phi_{\infty}}{k_B T}\right) \right] d\phi \end{aligned}$$

Change of variable

- After simple manipulation,
 - It can be simplified with $\bar{\phi} = \phi - \phi_{\infty}$:

$$\frac{d\phi}{dx} d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}} n_{int}$$

$$\sim \frac{n_{int}}{N_a} \times \left[\exp\left(-\frac{q\phi_{\infty}}{k_B T}\right) \left\{ \exp\left(-\frac{q\bar{\phi}}{k_B T}\right) - 1 \right\} - \exp\left(\frac{q\phi_{\infty}}{k_B T}\right) \left\{ \exp\left(\frac{q\bar{\phi}}{k_B T}\right) - 1 \right\} \right] d\bar{\phi} \sim \frac{N_a}{n_{int}}$$

Integration from 0 to ∞

- $\frac{d\phi}{dx}$ and $\bar{\phi}$ vanish at the substrate (∞).

– The integrated equation reads

$$\frac{1}{2} \left(\frac{d\phi}{dx} \Big|_0 \right)^2 = -\frac{q}{\epsilon_{si}} n_{int} \times \left[\exp \left(-\frac{q\phi_{\infty}}{k_B T} \right) \left\{ -\frac{k_B T}{q} \exp \left(-\frac{q\phi_s}{k_B T} \right) + \frac{k_B T}{q} - \phi_s \right\} - \exp \left(\frac{q\phi_{\infty}}{k_B T} \right) \left\{ \frac{k_B T}{q} \exp \left(\frac{q\phi_s}{k_B T} \right) - \frac{k_B T}{q} - \phi_s \right\} \right]$$

(Note that $\phi_s = \bar{\phi}(0)$)

Relation between ϕ_s and $d\phi/dx|_0$

- Then,

- Now, ϕ_s and $\frac{d\phi}{dx}|_0$ are related through

$$\left(\frac{d\phi}{dx}\right)_0^2 = \frac{2k_B T}{\epsilon_{si}} n_{int} \times \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\phi_s}{k_B T}\right) - 1 + \frac{q\phi_s}{k_B T} \right\} + \exp\left(\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(\frac{q\phi_s}{k_B T}\right) - 1 - \frac{q\phi_s}{k_B T} \right\} \right]$$

Draw $d\phi/dx|_0$ as a function of ϕ_s

- It is straightforward to draw it.

– $N_a = 4 \times 10^{15} \text{ cm}^{-3}$

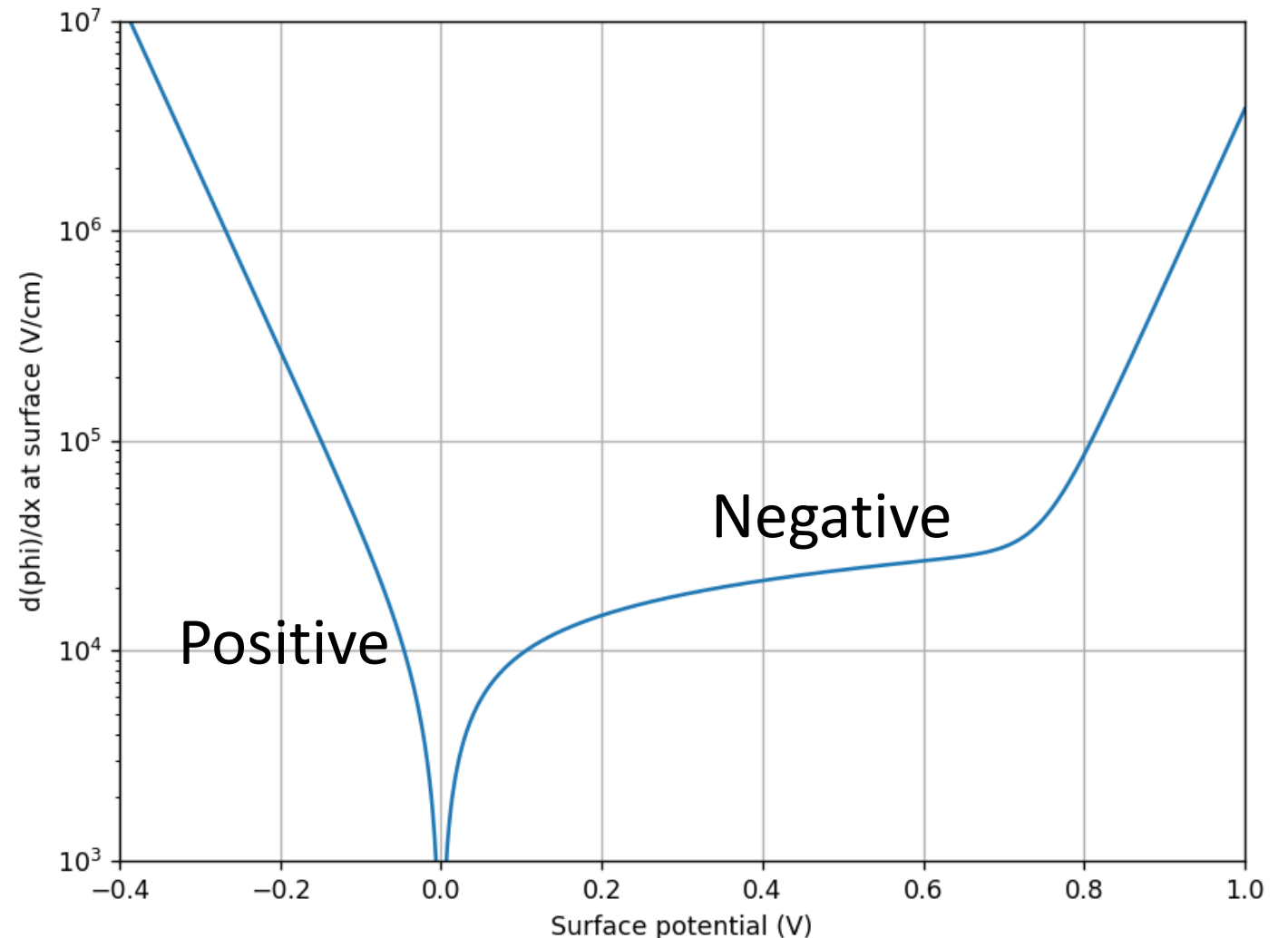
– Parameters

$q = 1.602192 \times 10^{-19} \text{ C}$

$n_{int} = 1.075 \times 10^{10} \text{ cm}^{-3}$

$k_B = 1.38065 \times 10^{-23} \text{ J/K}$

$T = 300 \text{ K}$

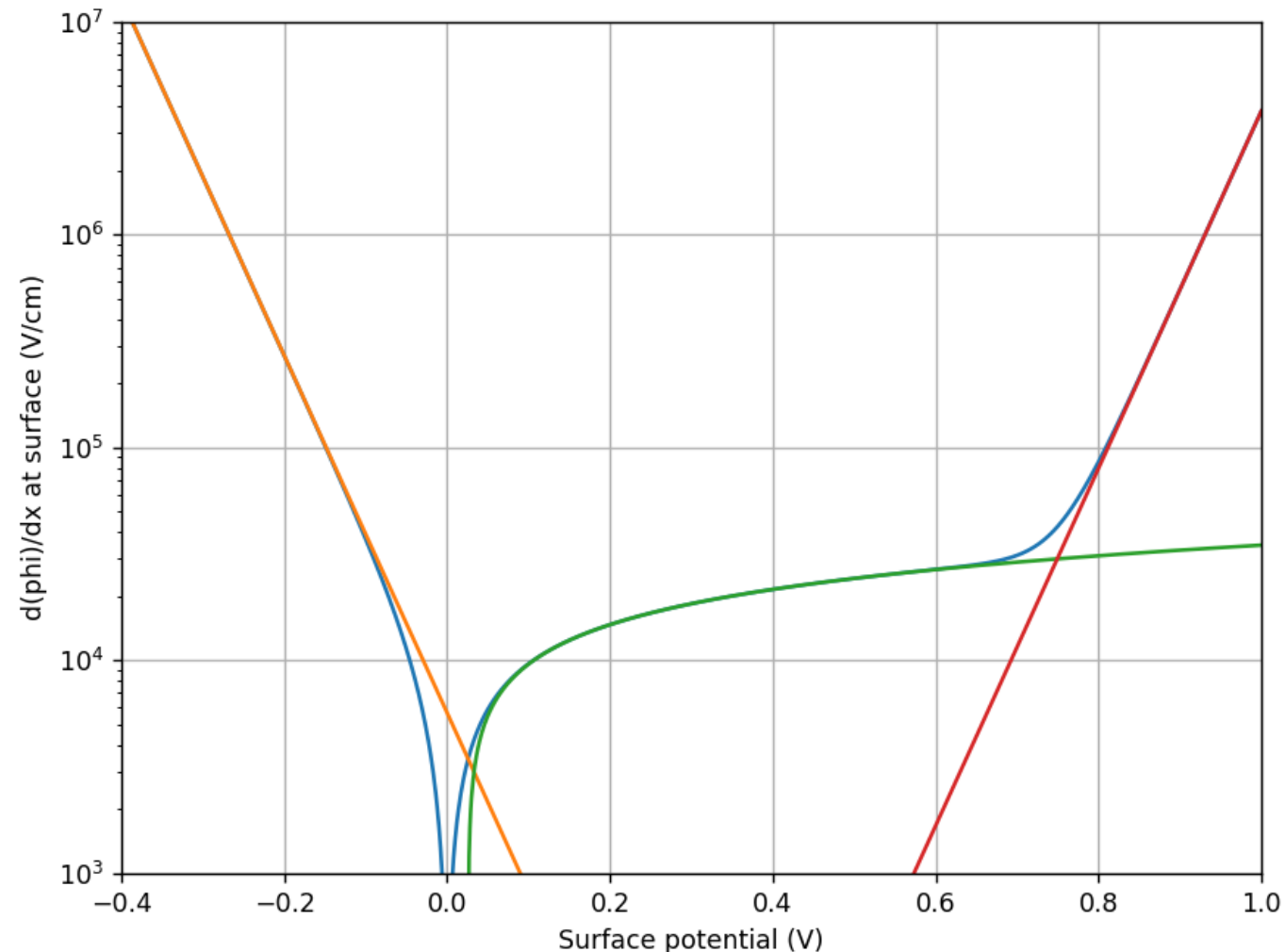


Approximate expressions

- Negative ϕ_s
$$\left(\frac{d\phi}{dx}\bigg|_0\right)^2 \approx \frac{2k_B T}{\epsilon_{si}} n_{int} \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \exp\left(-\frac{q\phi_s}{k_B T}\right) \right]$$
- Positive, but small ϕ_s
$$\left(\frac{d\phi}{dx}\bigg|_0\right)^2 \approx \frac{2k_B T}{\epsilon_{si}} n_{int} \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \frac{q\phi_s}{k_B T} \right]$$
- Sufficiently large ϕ_s
$$\left(\frac{d\phi}{dx}\bigg|_0\right)^2 \approx \frac{2k_B T}{\epsilon_{si}} n_{int} \left[\exp\left(\frac{q\phi_\infty}{k_B T}\right) \exp\left(\frac{q\phi_s}{k_B T}\right) \right]$$

Draw approximate expressions.

- Approximate expressions are reasonably good.



Calculate ϕ_s at a given $d\phi/dx|_0$

- For example, calculate ϕ_s corresponding to -1 MV/cm .
 - Residual function

$$\begin{aligned} &= \left(\left. \frac{d\phi}{dx} \right|_0 \right)^2 \\ &= -\frac{2k_B T}{\epsilon_{si}} n_{int} \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\phi_s}{k_B T}\right) - 1 + \frac{q\phi_s}{k_B T} \right\} \right. \\ &\quad \left. + \exp\left(\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(\frac{q\phi_s}{k_B T}\right) - 1 - \frac{q\phi_s}{k_B T} \right\} \right] \end{aligned}$$

Derivative

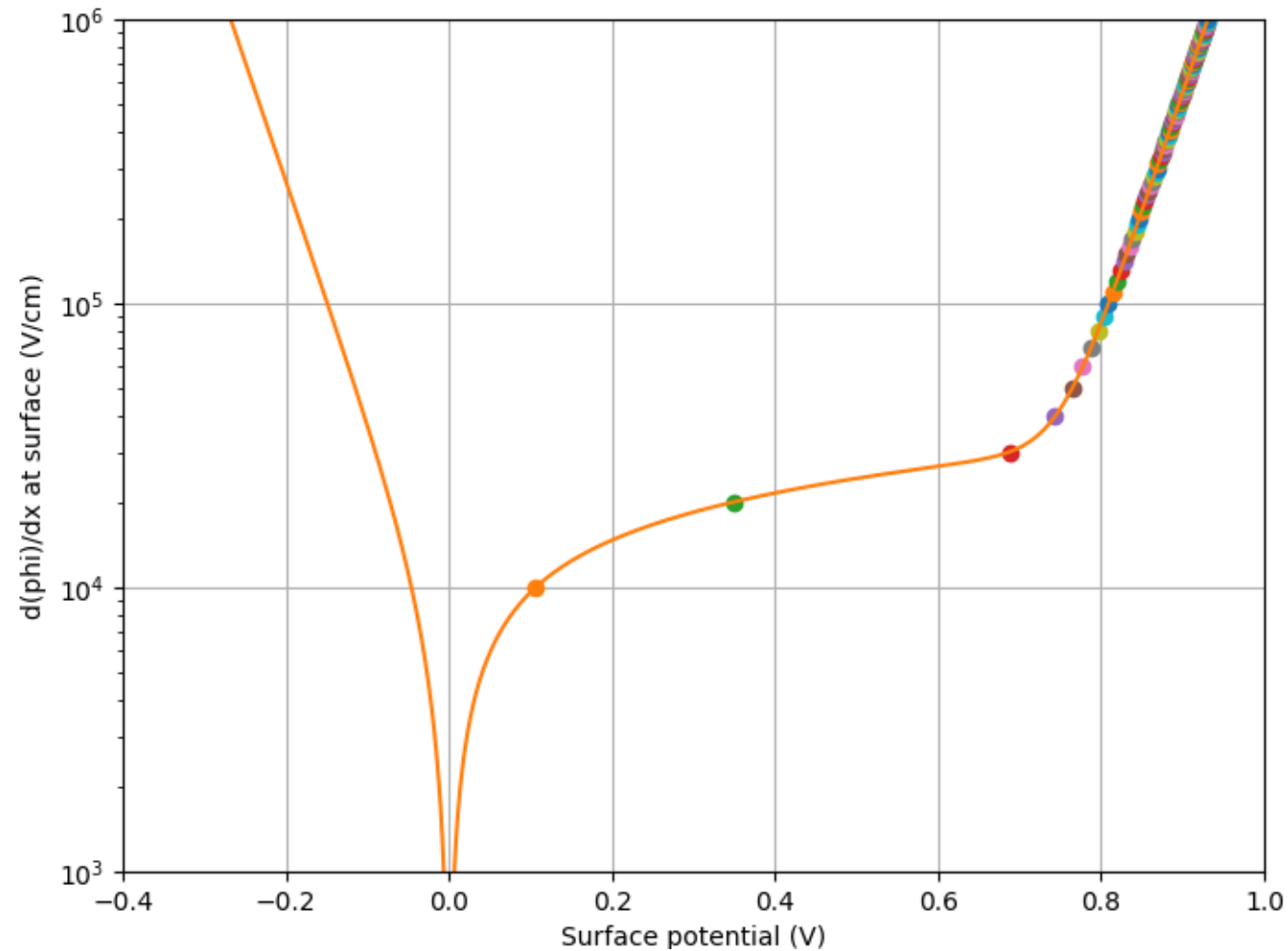
- For example, calculate ϕ_s corresponding to -1 MV/cm .
 - Derivative of the function

$$\frac{df}{d\phi_s} = -\frac{2q}{\epsilon_{si}} n_{int} \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \left\{ -\exp\left(-\frac{q\phi_s}{k_B T}\right) + 1 \right\} + \exp\left(\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(\frac{q\phi_s}{k_B T}\right) - 1 \right\} \right]$$

- We can implement the Newton method.
 - Once again, be careful about too large an update.
 - For example, the maximum update can be limited to $\frac{k_B T}{q}$.

Newton method

- Try your own initial guess.



Thank you!